Rewarding Learning

ADVANCED SUBSIDIARY (AS)
General Certificate of Education
2017

Digital Technology

Assessment Unit AS 2

assessing

Fundamentals of Digital Technology

[CODE]

SPECIMEN

MARK SCHEME

Version 2: 6 March 2017

			AVAILABLE MARKS
1	(a)	**kilobyte** 2^{10} or 1024 bytes [1] **Megabyte** 2^{20} or 1,048,576 bytes [1] [2]	
	(b)	**00001111_2** $1 + 2 + 4 + 8 = 15_{10}$ [1] + [1] **11110110_2** Invert 00001001_2 [1] Add 1 $00001010_2 = -(2 + 8) = -10_{10}$ [1] [4]	
	(c)	44_{10} [1] 00101100_2 [1] Invert 11010011_2 [1] Add 1 11010100_2 [1] [4]	
	(d)	**Indicative content** Candidates may discuss the following: 7 bits are used for characters [1] This gives a range of 128 characters [1] …which includes 96 displayable characters [1] …and 32 control codes [1]; A character is a letter or digit or special character [1] The 8th bit can be used as a parity bit [1] …for error checking [1] 8 bits can give a range of 256 [1]; Displayable characters; control codes [1] Any **five** points [5]	
	(e)	Unicode uses 16 bits [1] This provides a much greater range [1] This increases by a factor of 2^8 the number of different characters which can be represented [1] This is used to represent characters from most of the world's languages [1] Any **three** points [3]	18
2	(a)	**Data** Data is unprocessed [1] Example 45667 [1] [2] **Information** Processed data [1] Example 45667 is a ProductID [1] [2] **Knowledge** The application of information [1]	

			AVAILABLE MARKS
	Example Product 45667 needs re-ordering [1]	[2]	
(b)	Accuracy [1] Relevance [1] Up to date/currency [1] Completeness [1] Presentation [1] Reliability [1]		
	Any **two** points	[2]	

(c)		4	5	6	6	
	Weighting	5	4	3	2	[1]
	Product	20	20	18	12	[1]
	Sum	70				[1]
	Remainder	70 ÷ 11 = 6 R 4				[1] + [1]
	Check digit	11 − 4 = 7				[1]
						[6]

(d) Indicative content

Candidates may discuss the following:

Data validation will check that the date is a valid date [1]; range check will be used [1]
Example: the month number lies in the range 1 to 12 inclusive [1]
However, this does not check that the date is the one the user intended to enter [1]

Data verification confirms that the date entered was the intended date [1]
Example: day/month/year with reference to upper boundary [1]
day/month/year with reference to lower boundary [1]
Proof reading will be intended method [1]
The user will compare data entered with original [1]

Any **six** points [6]

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3 (a) Synchronises all internal processes/components [1] by generating pulses [1] at a constant rate [1] [3]

(b) The control unit [1] gets each instruction in sequence [1]
It decodes the instruction [1] and then carries it out by sending control signals to other parts of the computer [1] [3]

Any **three** points

(c) Each task has its own window [1]
Windows can be maximised and minimised [1]
Icons represent programs/files/shortcuts [1]

AVAILABLE MARKS

Icons can be double clicked to open [1]
Menus contain lists of options [1]
Menus can contain sub menus [1]
A pointer is used to move a cursor [1]
Rather than cursor key on a keyboard [1]

Any **four** points [4]

(d) A capacitive touch screen is used [1]
This consists of a grid of tiny, transparent electrodes [1]
The touch screen displays a keyboard [1]
When the screen is touched with a finger or stylus, there is a change in the electrostatic properties at that point [1]
This enables the typed character to be determined [1]

Any **four** points [4]

(e) Indicative content
Candidates may discuss the following:

A bitmap image retains all the image details
…at the expense of a large file size.
This can impact on the download time of the page.
The resolution of a JPEG image can be reduced
…using compression techniques.
This also reduces file size but at the expense of image quality.
GIF stands for Graphics Interchange Format and is one of the file formats usd to display indexed-colour graphics and images in HTML documents on the web. This means that a GIF will only display a maximum of 256 colours; as a result, GIFS are great for images with simple illustrations and blocks of colours, like logosw and icons, but probably not the best option for photographs.

Level of response	Marking criteria	Marks
Band 3: Excellent	The candidate accurately describes how an image can be stored using both of these methods. The candidate articulates clearly the advantages and disadvantages of each method referring directly to an image to be placed in a website. The candidate uses the appropriate Digital Technology vocabulary accurately. Presentation, spelling, punctuation and grammar are of a high standard.	[7]–[9]
Band 2: Good	The candidate describes how an image can be stored using both of these methods. The candidate identifies an advantage or disadvantage of each method but does not refer to an image to be placed in a website. The candidate uses some relevant Digital Technology vocabulary. Presentation, spelling, punctuation and grammar are sufficiently competent to make the response clear.	[4]–[6]
Band 1: Basic	The candidate identifies one feature of each type of image representation but does not refer to any advantage or disadvantage. The candidate makes limited use of Digital Technology vocabulary. Presentation, spelling, punctuation and grammar are such that the intended meaning is not clear.	[1]–[3]
The response is not worthy of credit		[0]

AVAILABLE MARKS

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4 **(a)** Memory management [1]
Allocation of resources such as printers [1]
Handling user input [1]
Handling output [1]

Any **three** points [3]

(b) **Data compression software**
Reduces the physical size of data [1] in order to save storage space [1]
Reduce transmission time [1]

Any **two** points [2]

Disk defragmenter software
Decreases access times [1] uses storage more efficiently [1] by merging fragmented files into one contiguous block or space [1]

Any **two** points [2]

(c) **Generic software**
Performs general tasks and is not limited to one particular application [1]
Software may not meet the exact user requirements [1]
Example: a spreadsheet application [1]

Any **two** points [2]

Special purpose software
Designed for a specific application [1]
Software will meet the exact user requirements [1]
Example: payroll software [1]

Any **two** points [2]

(d) Transactions are saved temporarily [1]
...until the end of the month [1]
All transactions for the month are used [1]
...to produce monthly bank statements [1]

Any **three** points [3]

(e) In real-time processing, each transaction is processed immediately [1]
Files are updated [1] before the next transaction takes place [1]
This is needed for ATM transactions as withdrawals and deposits to an account must be implemented immediately [1]
...so the account balance is always up to date [1]
Personal Identification Number (P.I.N.) changes updated immediately [1]

Any **four** points [4] 18

AVAILABLE MARKS

5 (a) The WWW

A system of interlinked hypertext documents [1] that are accessed via the Internet using a browser [1] [2]

The Internet

A global computer network [1] consisting of interconnected networks using communication protocols [1] [2]

URL

A unique address [1] identifying every resource on the Internet [1] [2]

(b) The Internet is a global network [1]
An intranet is a private network[1]; restricted network [1]; Requires a username and password [1]

Any **two** points [2]

(c) Holiday types: [1]
 [1]
Flights only [1]
Flights and hotel [1]
Cruises [1]

Click here to continue to CCEA Travel.com [1]

Any **four** points [4]

(d) Indicative content

Candidates may discuss the following:

The user’s browser performs any processing it can
... on the clients computer
... such as validate a debit/credit card number is numeric
Check all compulsory fields have been completed e.g. Expiry date, security number, name, credit card number.
Check all fields have been completed with valid data such as validating a debit/credit card number as numeric.
When the user clicks Proceed, the link with the server is re-established and the completed web page is sent to the server
The server completes the processing
...for example by authorising the payment details with the users bank
... authorising credit card details with bank
The server then sends the confirmation page back to the client computer.

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Level of response	Marking criteria	Marks
Band 3: Excellent	The candidate accurately and coherently articulates the difference between client-side and server-side processing in the specific context of the scenario. The candidate uses the appropriate Digital Technology vocabulary accurately. Presentation, spelling, punctuation and grammar are of a high standard.	[7]–[9]
Band 2: Good	The candidate describes the main difference between client-side and server-side processing but does not refer clearly to the scenario. The candidate uses some relevant Digital Technology vocabulary. Presentation, spelling, punctuation and grammar are sufficiently competent to make the response clear.	[4]–[6]
Band 1: Basic	The candidate attempts to distinguish between client-side and server-side processing. The candidate makes limited use of Digital Technology vocabulary. Presentation, spelling, punctuation and grammar are such that the intended meaning is not clear.	[1]–[3]
The response is not worthy of credit		[0]

Total **100**

AVAILABLE MARKS